

Rank-Deficiency and Coherence Analysis for Matrix-Completed CSI in mmWave MIMO Channels

Abstract—Low-rank matrix completion (LRMC) has emerged as a promising technique for channel state information (CSI) acquisition in millimeter-wave (mmWave) multiple-input multiple-output (MIMO) systems due to its ability to recover channel matrices from a limited number of observations. However, the effectiveness of matrix completion strongly depends on the structural properties of the underlying channel, particularly its rank and coherence. In this paper, a comparative analysis of matrix-completed CSI is conducted for four representative mmWave channel models: the One-Ring model, the Saleh–Valenzuela (SV) model, the Weichselberger model, and the QuaDRiGa mmMAGIC Urban Microcell Non-Line-of-Sight (UMi NLOS) model. An entropy-based effective rank metric is employed to quantify channel dimensionality, while singular-vector coherence is used to characterize channel recoverability. Channel reconstruction is performed using the classical Singular Value Thresholding (SVT) algorithm, and the minimum pilot ratio required to achieve a target normalized mean square error (NMSE) of is evaluated. The results show that increasing effective rank generally degrades reconstruction performance and increases pilot requirements. However, the realistic QuaDRiGa UMi NLOS channel exhibits poor recovery despite its low effective rank due to its high coherence, demonstrating that rank alone is insufficient to predict matrix completion performance. The study highlights the joint influence of rank and coherence on CSI recoverability and provides insights into the suitability of different mmWave channel models for low-rank matrix completion based channel estimation.

I. INTRODUCTION

Millimeter-wave (mmWave) communications have emerged as a key enabling technology for beyond-5G and 6G wireless systems due to the availability of large bandwidths capable of supporting multi-gigabit data rates. To compensate for the severe propagation losses encountered at mmWave frequencies, large-scale multiple-input multiple-output (MIMO) systems employing directional beamforming are commonly adopted. The performance of such systems, however, strongly depends on the availability of accurate channel state information (CSI). Conventional channel estimation techniques often require substantial pilot overhead, particularly for large antenna arrays, motivating the development of low-complexity CSI acquisition techniques.

Low-rank matrix completion (LRMC) has attracted considerable interest as an alternative framework for CSI estimation because it exploits the structural properties of the channel matrix rather than relying on explicit sparsifying dictionaries or array calibration information [1]. Under appropriate low-rank and incoherence conditions, matrix completion enables

accurate recovery of a channel matrix from a subset of observed entries, thereby reducing pilot overhead. The survey in [1] provides a comprehensive overview of matrix completion theory, including nuclear norm minimization and Singular Value Thresholding (SVT), which constitute the foundations of modern LRMC algorithms.

Motivated by the sparse scattering characteristics of mmWave propagation environments, several recent studies have applied matrix completion techniques to MIMO channel estimation. Rank-aware matrix completion methods were proposed in [2], [3] for fast time-varying mmWave channels, demonstrating improved reconstruction performance and reduced pilot requirements. Matrix completion has also been extended to extremely large antenna array (ELAA) systems through low-complexity recovery frameworks [4], while other works have exploited the low-dimensional structure of wireless channels for efficient CSI reconstruction and tracking [5]. Collectively, these studies demonstrate the potential of matrix completion as a practical tool for reducing CSI acquisition overhead in large-scale wireless systems.

Despite these advances, most existing works focus primarily on algorithm design and generally assume that practical mmWave channels inherently satisfy the low-rank and incoherence conditions required by matrix completion theory. While these assumptions are often justified by the limited number of dominant propagation paths and strong spatial correlation present in mmWave systems, their validity has rarely been examined systematically across different channel models. In particular, the quantitative relationship between channel rank, coherence, pilot requirements, and reconstruction performance remains insufficiently understood. Consequently, it is unclear whether low rank alone is sufficient to guarantee successful channel recovery or whether coherence can become the dominant factor governing recoverability.

Motivated by this gap, this work presents a comparative rank-deficiency analysis of matrix-completed CSI using four representative mmWave channel models: the One-Ring model [6], the Saleh–Valenzuela (SV) model [7], the Weichselberger model [8], and the QuaDRiGa [9] mmMAGIC Urban Microcell Non-Line-of-Sight (UMi NLOS) model. These models span a broad range of propagation conditions, from analytical correlation-based channels to realistic geometry-based stochastic channels. An entropy-based effective rank metric is employed to quantify channel dimensionality, while channel coherence is used to characterize the concentration of singular

vectors. Channel recovery is performed using the classical SVT algorithm [10], and the minimum pilot ratio required to achieve a target reconstruction accuracy is evaluated for each channel model.

The main contributions of this paper are summarized as follows:

- A comparative matrix completion framework is developed for the analysis of CSI recovery across One-Ring, Saleh–Valenzuela, Weichselberger, and QuaDRiGa channel models.
- The effective rank, coherence, and pilot requirements of different mmWave channel models are systematically characterized.
- The minimum pilot ratio required to achieve a target reconstruction accuracy is quantified using the classical Singular Value Thresholding (SVT) algorithm.
- The study demonstrates that successful matrix completion depends on the joint influence of rank and coherence rather than rank alone, providing insights into the recoverability of realistic mmWave MIMO channels.

II. CHANNEL MODEL

To investigate the impact of channel structure on matrix completion performance, four representative channel models are considered, namely the One-Ring model, the Saleh–Valenzuela (SV) model, the Weichselberger model, and the QuaDRiGa UMi NLOS model.

1) *One-Ring Channel Model*: The One-Ring model describes a propagation environment in which the receiver is surrounded by local scatterers while the transmitter is located sufficiently far away. Under this assumption, the spatial correlation between antenna elements is governed by the angular spread of the arriving waves. For a uniform linear array (ULA), the correlation matrix is approximated as

$$[\mathbf{R}]_{m,n} = J_0(2\pi d|m-n|\Delta), \quad (1)$$

where $J_0(\cdot)$ denotes the zeroth-order Bessel function of the first kind, d is the antenna spacing normalized by wavelength, and Δ denotes the angular spread.

The correlated MIMO channel is generated as

$$\mathbf{H} = \mathbf{R}_r^{1/2} \mathbf{H}_w \mathbf{R}_t^{1/2}, \quad (2)$$

where $\mathbf{H}_w$ is an i.i.d. complex Gaussian matrix and $\mathbf{R}_r$ and $\mathbf{R}_t$ represent the receive and transmit correlation matrices, respectively.

2) *Saleh–Valenzuela Channel Model*: The Saleh–Valenzuela (SV) model captures the sparse and clustered nature of mmWave propagation. The channel matrix is represented as

$$\mathbf{H} = \sqrt{\frac{N_t N_r}{N_{cl} N_{ray}}} \sum_{c=1}^{N_{cl}} \sum_{r=1}^{N_{ray}} \alpha_{c,r} \mathbf{a}_r(\theta_{c,r}^{AoA}) \mathbf{a}_t^H(\theta_{c,r}^{AoD}), \quad (3)$$

where N_{cl} denotes the number of scattering clusters, N_{ray} is the number of rays per cluster, $\alpha_{c,r}$ is the complex path gain,

and $\mathbf{a}_r(\cdot)$ and $\mathbf{a}_t(\cdot)$ denote the receive and transmit steering vectors, respectively.

3) *Weichselberger Channel Model*: The Weichselberger model provides a generalized framework for correlated MIMO channels by explicitly modeling the coupling between transmit and receive eigenmodes. The channel matrix is expressed as

$$\mathbf{H} = \mathbf{U}_r (\mathbf{\Omega} \odot \mathbf{G}) \mathbf{U}_t^H, \quad (4)$$

where $\mathbf{U}_r$ and $\mathbf{U}_t$ denote the receive and transmit eigenvector matrices, $\mathbf{G}$ is an i.i.d. complex Gaussian matrix, $\odot$ denotes the Hadamard product, and $\mathbf{\Omega}$ is the power coupling matrix describing the coupling strength between transmit and receive eigenmodes.

a) *Coupling Scenarios*: To investigate the influence of eigenmode coupling on matrix completion performance, four representative power coupling structures are considered:

- **Case 1 (Sparse Coupling)**: Channel power is concentrated within a small subset of eigenmode pairs, resulting in a highly rank-deficient channel.
- **Case 2 (Diagonal Coupling)**: Power is concentrated along the main diagonal of $\mathbf{\Omega}$, representing nearly independent spatial modes.
- **Case 3 (Exponential Coupling)**: Coupling strength decreases exponentially away from the diagonal, modeling moderate scattering environments.
- **Case 4 (Uniform Coupling)**: Power is distributed across all eigenmode pairs, corresponding to rich scattering conditions and higher channel rank.

These coupling structures provide different levels of rank deficiency and coherence, enabling the evaluation of their impact on matrix completion performance.

In this work, a rank-constrained coupling matrix is employed to generate channels with controlled rank deficiency and different eigenmode coupling characteristics.

4) *QuaDRiGa UMi NLOS Channel Model*: To evaluate matrix completion performance under realistic propagation conditions, channels are generated using the QuaDRiGa simulator under the mmMAGIC Urban Microcell Non-Line-of-Sight (UMi NLOS) scenario. QuaDRiGa follows a geometry-based stochastic channel modeling approach in which the channel response is represented as

$$\mathbf{H}(f) = \sum_{n=1}^{N_p} \alpha_n e^{-j2\pi f \tau_n} \mathbf{a}_r(\phi_n) \mathbf{a}_t^H(\theta_n), \quad (5)$$

where N_p denotes the number of propagation paths, α_n is the path gain, τ_n is the path delay, and ϕ_n and θ_n represent the angles of arrival and departure, respectively.

Unlike analytical channel models, QuaDRiGa incorporates realistic propagation effects such as cluster evolution, delay spread, angular spread, and spatial consistency, thereby providing a practical benchmark for evaluating matrix completion performance in mmWave MIMO systems.

TABLE I: Common Simulation Parameters

Parameter	Value
Number of transmit antennas (N_t)	8
Number of receive antennas (N_r)	16
Carrier frequency (f_c)	28 GHz
Antenna spacing	0.5λ
Monte Carlo realizations	50
Target NMSE	-15 dB
SVT threshold (τ)	$5 \max(N_t, N_r)$
SVT step size (δ)	1.2
Convergence tolerance	10^{-4}

TABLE II: Model-Specific Parameters

Model	Parameter	Value
One-Ring	Angular Spread	0.5° - 20°
	Mean Angle	30°
SV	Number of Clusters	1-8
	Rays per Cluster (N_{ray})	3
	Angular Spread	10°
Weichselberger	Desired Rank (R)	4
	Scenarios	1-4
QuaDRiGa UMi NLOS	Number of Clusters	2-10
	Bandwidth	100 MHz

III. EXPERIMENTAL SETUP

A comparative study of matrix completion performance was conducted using four representative MIMO channel models: the One-Ring model, the Saleh-Valenzuela (SV) model, the Weichselberger model, and the QuaDRiGa UMi NLOS model. All simulations were performed for a 16×8 MIMO system, where $N_r = 16$ and $N_t = 8$. A carrier frequency of 28 GHz was considered, and results were averaged over 50 independent Monte Carlo channel realizations.

For the One-Ring model, spatial correlation was controlled through the angular spread parameter, which was varied. The angular spread was varied over 0.5° , 2° , 5° , 10° , and 20° . The mean angle was fixed at 30° and the antenna spacing was set to 0.5λ . The channel matrix was generated by applying transmit and receive correlation matrices to an i.i.d. Rayleigh fading channel.

For the SV model, clustered mmWave propagation was considered. The number of scattering clusters was varied. For the SV model, the number of scattering clusters was varied over 1, 2, 3, 5, and 8, while the number of rays per cluster was fixed at $N_{\text{ray}} = 3$. The angular spread was fixed at 10° , and the antenna spacing was maintained at 0.5λ .

The Weichselberger channel model was generated using a rank-constrained coupling matrix with desired rank $R = 4$. Four different coupling scenarios were investigated to evaluate the impact of eigenmode coupling on channel recoverability.

For the QuaDRiGa simulations, the mmMAGIC Urban Microcell Non-Line-of-Sight (UMi NLOS) scenario was employed. The number of clusters was varied as 2, 4, 6, 8, and 10. The transmitter and receiver were equipped with 8-element and 16-element 3GPP mmWave antenna arrays, respectively. The base station height was fixed at 10 m, while the user equipment height was set to 1.5 m. The receiver was located

TABLE III: One-Ring Model: Effect of Angular Spread on Effective Rank, Coherence, and Required Pilot Ratio

Angular Spread (deg)	Effective Rank	Coherence	Required Pilot Ratio
0.5	1.0617	13.9440	0.39
2	1.2972	12.1400	0.41
5	1.7452	9.1703	0.51
10	2.4003	6.5880	0.62
20	3.6306	4.1466	0.76

50 m from the transmitter. Channel frequency responses were generated over a bandwidth of 100 MHz using 128 subcarriers, and the channel matrix corresponding to the central subcarrier was extracted for analysis.

For each channel realization, the singular value decomposition (SVD) of the channel matrix was computed. The entropy-based effective rank was evaluated as

$$r_{\text{eff}} = \exp \left(- \sum_i p_i \log(p_i) \right), \quad (6)$$

where

$$p_i = \frac{\sigma_i}{\sum_j \sigma_j}, \quad (7)$$

and σ_i denotes the i -th singular value. Channel coherence was computed using the singular-vector coherence metric.

Channel recovery was performed using the classical Singular Value Thresholding (SVT) algorithm. The threshold parameter was selected as

$$\tau = 5 \max(N_r, N_t), \quad (8)$$

with step size $\delta = 1.2$, maximum iteration count of 200, and convergence tolerance of 10^{-4} .

To determine the minimum pilot overhead required for successful channel recovery, a binary-search procedure was employed. A recovery was declared successful when at least 70% of the trials achieved the target NMSE of -15 dB.

The reconstruction accuracy was quantified using the normalized mean square error (NMSE),

$$\text{NMSE} = \frac{\|\mathbf{H} - \hat{\mathbf{H}}\|_F^2}{\|\mathbf{H}\|_F^2}, \quad (9)$$

where $\mathbf{H}$ and $\hat{\mathbf{H}}$ denote the original and reconstructed channel matrices, respectively. Finally, the effective rank, coherence, required pilot ratio, required number of observations, and achieved NMSE were averaged across all Monte Carlo realizations and used to evaluate the relationship between channel structure and matrix completion performance.

IV. RESULTS

Tables III-VII summarize the effective rank, coherence, and required pilot ratio for the considered channel models. For the One-Ring model, increasing the angular spread from 0.5 to 20 increases the effective rank from 1.06 to 3.63 while reducing the coherence from 13.94 to 4.15. Although lower

TABLE IV: Saleh–Valenzuela (SV) Model: Effect of Number of Clusters on Effective Rank, Coherence, and Required Pilot Ratio

Number of Clusters	Effective Rank	Coherence	Required Pilot Ratio
1	2.1900	6.7816	0.60
2	3.5309	3.8183	0.73
3	4.2942	2.7455	0.77
5	5.1141	2.1579	0.79
8	5.4133	2.0701	0.79

TABLE V: Weichselberger Model: Effect of Coupling Scenarios on Effective Rank, Coherence, and Required Pilot Ratio

Scenario	Effective Rank	Coherence	Required Pilot Ratio
1	3.0496	4.8497	0.74
2	3.1327	4.6723	0.75
3	3.0773	4.8127	0.75
4	3.0415	4.8468	0.74

TABLE VI: QuaDRiGa UMi NLOS

Number of Clusters	Effective Rank	Coherence	Required Pilot Ratio
2	1.3442	11.072	0.54
4	1.3453	10.788	0.55
6	1.3735	10.524	0.56
8	1.3427	10.739	0.55
10	1.3789	10.683	0.55

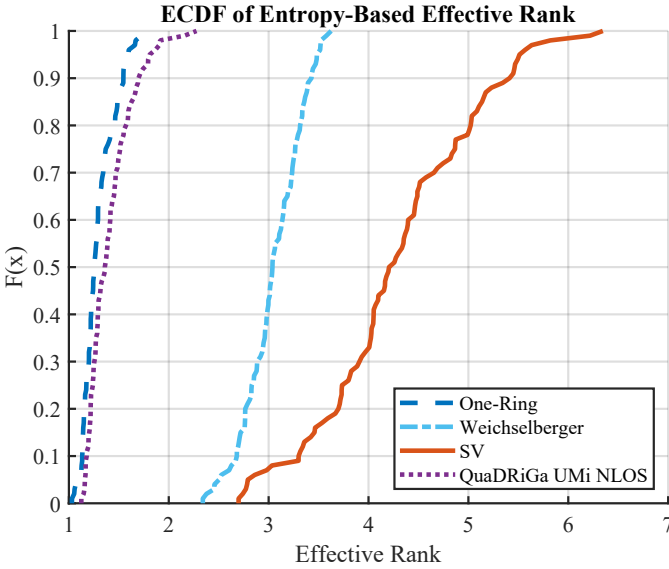


Fig. 1: ECDF of entropy-based effective rank for different channel models. The SV model exhibits the highest effective rank, while the One-Ring and QuaDRiGa UMi NLOS models show lower effective rank distributions.

coherence generally favors matrix completion, the increase in rank results in a higher pilot requirement, demonstrating the trade-off between rank and incoherence.

For the Saleh–Valenzuela (SV) model, increasing the number of clusters increases the effective rank from 2.19 to 5.41 and reduces the coherence from 6.78 to 2.07. The richer scat-

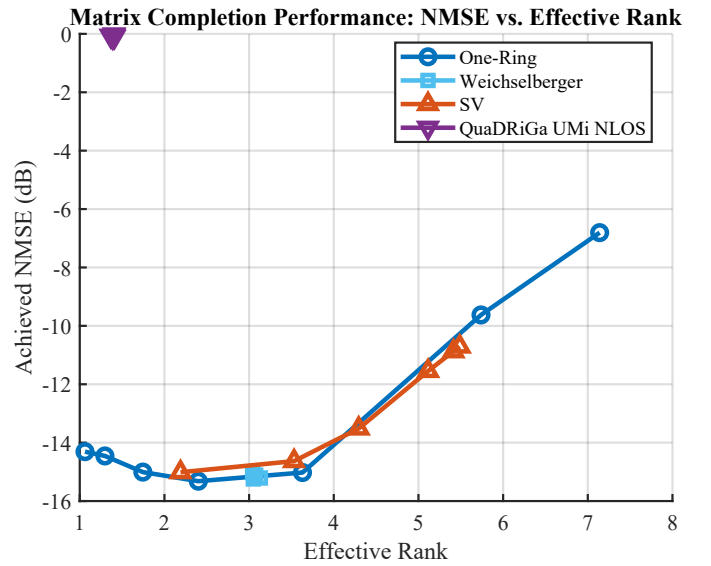


Fig. 2: Achieved NMSE as a function of effective rank for different channel models.

tering environment therefore produces a higher-rank channel, leading to an increase in the required pilot ratio. This indicates that the rank growth dominates the benefits obtained from reduced coherence.

The Weichselberger model exhibits relatively stable characteristics across different coupling scenarios, with an effective rank of approximately 3.1 and coherence around 4.8. Consequently, the required pilot ratio remains nearly constant, indicating that moderate variations in the coupling matrix have limited impact on the matrix completion requirements.

In contrast, the QuaDRiGa UMi NLOS model maintains a very low effective rank of approximately 1.35 across all cluster configurations. However, the coherence remains high (around 10.5–11), indicating strong concentration of singular vectors. As a result, the channel remains difficult to recover despite its pronounced rank deficiency.

Fig. 1 presents the empirical cumulative distribution function (ECDF) of the entropy-based effective rank. The SV model exhibits the highest effective-rank distribution, followed by the Weichselberger model. The One-Ring and QuaDRiGa UMi NLOS models remain concentrated at lower effective-rank values, reflecting their stronger spatial correlation and reduced channel dimensionality.

Fig. 2 illustrates the relationship between effective rank and achieved NMSE for the considered channel models. In general, the reconstruction error increases as the effective rank increases, indicating that channel recovery becomes more challenging when the number of significant spatial dimensions grows. This trend is particularly evident for the One-Ring and SV models.

For the One-Ring model, the effective rank increases from approximately 1.06 to 7.14 as the angular spread increases. At low angular spreads, the channel energy is concentrated within a few dominant eigenmodes, resulting in a highly rank-

deficient channel that can be accurately reconstructed using matrix completion. Consequently, the achieved NMSE remains close to the target value of (-15) dB. As the angular spread increases, additional spatial dimensions become significant, leading to a gradual increase in effective rank and a corresponding degradation in reconstruction performance. The NMSE eventually rises to approximately (-7) dB, demonstrating the increased sampling complexity associated with higher-rank channels.

A similar behavior is observed for the SV model. Increasing the number of scattering clusters introduces additional propagation paths, causing the effective rank to increase from approximately 2.19 to 5.49. Although the coherence decreases with increasing cluster count, the increase in rank dominates the reconstruction process, resulting in a gradual degradation in NMSE from approximately (-15) dB to (-11) dB. Compared with the One-Ring model, however, the SV model exhibits a slower degradation rate due to its lower coherence values.

The Weichselberger model occupies an intermediate region in the figure, maintaining an effective rank close to 3.1 and an NMSE around (-15) dB. Since both the effective rank and coherence remain nearly unchanged across the considered coupling scenarios, the reconstruction performance is also relatively stable. This indicates that moderate variations in the eigenmode coupling structure do not significantly affect channel recoverability.

The most notable observation is the performance of the QuaDRiGa UMi NLOS model. Despite exhibiting the lowest effective rank among all considered channels, its NMSE remains close to 0 dB, indicating poor channel reconstruction. This result appears counterintuitive because low-rank channels are generally expected to be easier to recover. However, the QuaDRiGa channel also exhibits very high coherence, implying that the channel energy is concentrated within a small subset of singular vectors. Such a structure violates the incoherence assumptions required by matrix completion theory and significantly reduces the effectiveness of the SVT algorithm. Consequently, the channel remains difficult to reconstruct despite its strong rank deficiency.

Overall, the results demonstrate that effective rank alone is insufficient to characterize matrix completion performance. While increasing rank generally degrades reconstruction accuracy, the poor performance of the QuaDRiGa UMi NLOS model confirms that coherence plays an equally important role in determining recoverability. Therefore, successful matrix completion requires both a low-rank structure and sufficiently low coherence, highlighting the need to jointly consider these two metrics when evaluating channel reconstruction performance in mmWave MIMO systems.

V. CONCLUSION

This paper presented a comparative study of matrix completion performance for four representative mmWave MIMO channel models, namely the One-Ring, Saleh–Valenzuela (SV), Weichselberger, and QuaDRiGa UMi NLOS models.

Using the classical Singular Value Thresholding (SVT) algorithm, the relationships among effective rank, coherence, pilot requirements, and reconstruction accuracy were systematically investigated.

The results showed that increasing angular spread in the One-Ring model and increasing the number of clusters in the SV model both lead to higher effective rank, which increases the pilot ratio required for accurate channel recovery and degrades the achieved NMSE. The Weichselberger model exhibited relatively stable effective rank and coherence across different coupling scenarios, resulting in consistent reconstruction performance. In contrast, the QuaDRiGa UMi NLOS model demonstrated that low rank alone does not guarantee successful matrix completion. Despite exhibiting the lowest effective rank among the considered models, its high coherence resulted in poor reconstruction accuracy, highlighting the limitations of relying solely on rank-based metrics.

Overall, the study confirms that matrix completion performance is jointly governed by both effective rank and coherence. While rank determines the intrinsic dimensionality of the channel, coherence influences the distribution of information across channel entries and significantly affects recoverability. Therefore, accurate assessment of CSI reconstruction performance requires the simultaneous consideration of these two metrics rather than rank alone.

Future work will focus on extending the analysis to larger massive MIMO configurations, additional standardized channel models, and coherence-aware sampling and matrix completion techniques to improve CSI recovery in highly coherent propagation environments.

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